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Aircraft Hangar Door Project for RCDL @ Ahemdabad Airport



Quotation

ASSA ABLOY

ASSA ABLOY Entrance Systems

The global leader in door opening solutions

Technical Details: Model VL3190 (Hangar Door)

Quotation reference: **SQT/19-20/02169R**

Date: 18th Oct 2019

Mahimtura Consultants Private Limited
Unique House, 3rd Floor,
25 S.A. Brelvi Road, Fort,
Mumbai – 400001, INDIA

Kind Attn: Mr. Prashant Tidke
Contact No: 9833482936
Email: prashant@mahimtura.net

Project: **Hangar Door Project for RCDL @ Ahemdabad Airport**

Dear Sir,

Thank you for giving us an opportunity to participate with our offer on the above project.

By choosing ASSA ABLOY Entrance Systems as the supplier of your door automation products, you will receive the following advantages:

- You are working with a Global Provider with financial strength and over 50 years' experience of the global market place in Entrance Solutions.
- World leading products ensure excellent reliability and life-cycle costs to give you peace of mind.
- Project support for both specification development and design suitability
- A comprehensive product helping to provide value engineered solutions

Major Benefits using Megadoors

- No floor track (and therefore no concrete grade beam) needed; a substantial saving in civil cost.
- Faster opening and closing time due to vertical lifting, resulting in a substantial reduction in loss of energy.
- Minimal side-room required (no "door pockets"); a substantial saving in space and cost of land, this is especially relevant, when a series of hangars are built adjacent to each other.
- The doors have very tight sealing, to protect against water, sand and dust ingress, which is essential for a maintenancehangar.

We wish you success with your project and hope that we may be part of your current & future projects as a solution provider. We will follow up this quotation with a call to you within the next few days.

A. Requirement

Hangar purpose, type of aircraft

The hangar will be used for the storage of Aircraft. The hangar door must provide a tight seal to reduce energy loss and provide protection against ingress of dust, sand and water. A short opening and closing time of the door will contribute to the saving of energy.

The clear opening: **(CW) 67 Mtr X (CH) 14.7 Mtr /14 meter clear under mullion**

PART 1 -- GENERAL

It is the intent of this specification to identify the nature of the required door and to define the minimum standards for the quality, construction, and operation of the required door. Minor variations, in accordance with standard practice, shall be indicated on shop drawings and submitted for approval.

1.1 CODES AND STANDARDS

- The following standards form a part of this specification to the extent referenced. References in the body of this specification are by basic designation only.
- TUV Certification (Certification of Safety Arrestors)
- DIN, BS and ASTM Norms

PART 2 – Technical Specification

VERTICAL LIFTING FABRIC DOOR SYSTEM:

A. **General**: Vertical Lifting Fabric Doors door shall be designed in such a way that the material, construction, and operation of the door shall conform to the performance specifications contained herein and it doesn't consist of any protruding parts affecting personal safety in site. Doors shall operate properly without binding, interference, or damage to the adjacent structure

B. **Door Leaf:** The door leaf shall be made of two layers of very strongly vinyl coated polyester fabric weighing approximately 700g/sq.m), capable of carrying 2400N/5cm. It shall be UV stabilised, self extinguishing (0 -25 flame spread), and suitable to withstand temperatures between – -30°C to +70°C. The translucent material if used, shall also meet the same requirements and loading as the standard door fabric. The fabric shall be attached to both sides of the intermediate beams, top beam and bottom beam with self tapping screws through corrosion resistant aluminium batten strips Powder coated clamp strips to match the color of the fabric. The door fabric shall be pulled tight between

the intermediate beams when the door is fully closed and wind lock engaged. The fabric shall have been tested to meet the criteria of attached test documents and standards as below :

- **Flame retardency:** tested according: EN 13501-1, B-s2,d0.
- **Tensile strength:** 2500/2000 N / 5 cm acc to EN ISO 1421-1
- **Tear resistance:** 400 / 300 N acc to DIN 53363
- **Light resistance:** ≥ 7 acc to ISO 105-B02
- **Radar reflecting coefficient:** 0,3dB, ~0,1%

C. **Door Color:** shall be a manufacturer's standard color, Red, Blue, Grey, White, White aluminum, Tan, Green, Graphite or Translucent.

D. **Doors:** shall consist of hoist up fabric doors with intermediate aluminium beams or trusses. Fabric shall be gathered above the head of the opening. Regardless of location, the minimum design wind load shall be only calculated to 1.1 kPa. Doors shall be fully operable during a wind speed at 20m/sec.

E. **Door Beams:** The intermediate beams shall be corrosion resistant extruded aluminium tempered after the extrusion process to ensure required strength and have a length to depth ratio not to exceed 100:1 to meet the required wind load. They shall be spaced 700 to 1400 metres apart, dependent upon the wind load. At each end of the beams there shall be a guide block of self-lubricating nylon material, or rollers that run along the guides. The intermediate beams must have a minimum depth of 290mm. The intermediate door beams and guide block or rollers shall be designed to carry the full design wind load without failure or being pulled from the door guides. There shall be no multiple lifting cables and holes distributed along the length of the intermediate beam. Steel intermediate beams will not be accepted.

F. **Bottom Section:** The bottom section is to be made of steel. A rubber seal to be fixed to the section to ensure tightness against the floor /ground. Bottom section should prevent lifting of door during wind load and secured at locked position with an automatic wind lock system.

G. **Door Guide:** The vertical guides shall be an integral part of the door, made of extruded aluminium with a suitable depth and width dependent on the size of the intermediate beams. The guides shall be designed to provide weather sealing on the inside and the outside faces. The guides shall be designed to provide sufficient strength to transfer the wind load and safety arrestor forces from the door beams, at maximum wind load, to the jamb steel support system. The guides shall be easily replaceable in case of damage. The installer shall furnish jamb structural steel fasteners of size and type required for attachment of the guide rails as shown on the approved shop drawings. The door guides must be manufactured with a centre T shaped extruded aluminium section, which prevents the intermediate beams leaving the guides in extreme wind conditions and two aluminium side seal extrusions. Steel guides are not acceptable.

H. **Drive Units:** The door leaf shall have an IEE / EN labelled single or dual motor drive system with torque sized as appropriate for the weight of the door leaf. The gear motor shall be equipped with a drum on which the belts/ropes are wound. If a single motor drive is used, the belts/ropes shall be wound on the same drum. The belts/ropes shall be

attached to the bottom beam via the safety arrestors. The gear motor shall be removable without disturbing the limit switch adjustment. Drive units have to be manufactured in USA, Europe or UK.

I. **Belt System:** A maximum of two belts/ropes per door, running inside the doors guides, shall be used to transmit motive force to the door unit. Similarly, the quantity of sheaves used to guide the belts/ropes shall be minimised to reduce maintenance requirements and spare parts inventories. Belts/ropes shall be installed free of any kinks and the system design and sheave diameter shall be such to prevent the occurrence of any kinks or abnormal stress in the operating belts/ropes. Where belts/ropes pass through openings in the building structure the openings shall be constructed so as to prevent abrasion, wear, or damage to the belts/ropes. Sheave units shall be installed in accessible locations that allow inspection and preventive maintenance. Sheaves shall not be located in enclosed locations that are not readily accessible for visual inspection.

J. **Safety Arrestors:** Each door leaf shall be supplied with two safety arrestors that activate and support the door in the event a belt/rope should break. Each safety arrestor type shall be tested to the standards established by TUV-control for the required door leaf weight and hold a valid certificate. A copy of the certification shall be a requirement of the submittals. The movable mullions, when provided, shall be provided with a lifting motor that includes back up arresting system to prevent the mullion from falling in case of chain or rope rupture. Third party TUV certified certificate for the safety arrestor to be provided. Only TUV tested and certified certificate shall be acceptable and evidence needs to be provided.

K. **SlackbeltBreaker:** A safety device shall be used on all door leaf belts/ropes which will sense a slack belt/rope condition and cut power to the appropriate drive unit to prevent an unsafe condition.

L. **Bottom Beam:** The bottom beam shall be designed with a suitable width and depth to carry the load of the intermediate beams when the door is open, and to ensure full closing and a tight floor seal in heavy winds. A heavy U-shaped bottom rubber seal shall be provided to form a tight seal with the floor, even on uneven surfaces.

M. **Self Activating Wind Lock:** Each door leaf shall be provided with two self activating wind locks which activate and lock the bottom beam when the door reaches its closed position. This locking action shall act to maintain a tight floor seal and intermediate stability even under conditions of heavy wind. A proximity switch, with indicator shall be provided to notify the door operator that all wind locks are engaged properly.

N. **Header Box:** Header boxes, when provided, shall be constructed of carbon steel and factory painted to C3 environmental standards with rust inhibitive paint.

O. **Control Panel:** The door manufacturer shall supply control panels for installation by the electrical subcontractor to operate the door units. The control panel shall also contain interlocks to preclude personnel injury, including interlock between the power supply system and use of the hand crank for manual operation of the door unit. The control panel shall be mounted a minimum of 1200mm above the floor. The control panel shall be IP 65 rated. The control panel will be a PLC based system. In case of multiple part door there must be separate control units for each door which communicates with the master control panel.

P. **Swing up Mullion:** The swing-up mullions shall be designed to swing up under the door leaf in the raised position. They shall be constructed of steel and painted to fulfil environmental Class C3 according to ISO 12944-2. The mullion hinge pivot shall be of bearing design. The mullion hinge pivot shall be of a maintenance free bearing design. The electrical control panel shall provide an interlock function to coordinate door leaf and mullion operation operation to prevent equipment damage. Mullion push buttons for both up and down operation shall be of the constant pressure type. The mullion must open/close automatically directly from the control panel. The mullion should not require any manually locking in closed position. The mullion must have flexible hinge design in multiple directions to cater to higher deflections from the roof truss.

Q. **Wind Resistance:** Doors shall be engineered for a wind load of 1.1 kPA.

R. **Water penetration:** Doors shall be engineered in accordance to EN12425 Class 3 with resistance to water penetration of 110PA for a closed door with or without pass door. The door shall be exceptionally designed to withstand water spray for >20minutes under pressure of >50PA (N/m^2).

S. **Air Tightness:** Doors shall be engineered in accordance to EN 12426 **Class 2- 3,** with or without pass-door.

If you have any questions after reading this proposal, or if you need more information, please do not hesitate to contact us.

Kind Regards,
Ms. Varsha
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